

3(a)	anabolic / condensation / polymerisation ;	1
3(b)	1 to, phosphorylate / add phosphate to, glucose ; <i>and any one from:</i> 2 stops glucose leaving the cell ; 3 activates glucose / makes glucose less stable ; 4 to make fructose (1,6) bi(s)phosphate ;	2
3(c)	cytoplasm and mitochondrial matrix ;	1

Question	Answer	Marks																					
3(d)	any two from: 1 decarboxylated / loses carbon dioxide ; 2 dehydrogenated / oxidised / loses H ; 3 →, 2C / acetyl, (group) joins coenzyme A to make acetyl coA ;	2																					
3(e)	any three from: <table border="1"> <thead> <tr> <th></th><th>mitochondria</th><th>chloroplasts</th></tr> </thead> <tbody> <tr> <td>1</td><td>oxidative phosphorylation</td><td>photophosphorylation ;</td></tr> <tr> <td>2</td><td>inner mitochondrial membrane / crista(e)</td><td>thylakoid membrane ;</td></tr> <tr> <td>3</td><td>reduced NAD / reduced FAD, give e⁻ / H⁺</td><td>photolysis / water / PS1 / chlorophyll, give e⁻ / H⁺ ;</td></tr> <tr> <td>4</td><td>(H⁺ →) intermembrane space</td><td>(H⁺ →) thylakoid, space / lumen ;</td></tr> <tr> <td>5</td><td>oxygen, final, e⁻ / H⁺, acceptor</td><td>NADP final, e⁻ / H⁺, acceptor ;</td></tr> <tr> <td>6</td><td>(makes) water / H₂O</td><td>(makes) reduced NADP ;</td></tr> </tbody> </table>		mitochondria	chloroplasts	1	oxidative phosphorylation	photophosphorylation ;	2	inner mitochondrial membrane / crista(e)	thylakoid membrane ;	3	reduced NAD / reduced FAD, give e ⁻ / H ⁺	photolysis / water / PS1 / chlorophyll, give e ⁻ / H ⁺ ;	4	(H ⁺ →) intermembrane space	(H ⁺ →) thylakoid, space / lumen ;	5	oxygen, final, e ⁻ / H ⁺ , acceptor	NADP final, e ⁻ / H ⁺ , acceptor ;	6	(makes) water / H ₂ O	(makes) reduced NADP ;	3
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